1. Problem Definition

Modern text classifiers can achieve high accuracy on topic classification, but accuracy alone does not explain what the model has actually learned. A classifier may appear successful while relying on shallow cues such as source markers, named entities, repeated keywords, or short title patterns instead of understanding the broader context of an article. This is a serious problem because models that depend on shortcuts may fail when the input is noisy, ambiguous, or slightly modified. Our project addresses this issue by studying what different text classification models rely on when classifying news articles. The central question of the project is: what do text classifiers learn from the title and description of a news article, and how robust are those learned patterns?

We will focus on AG News topic classification. This dataset is suitable for the problem because each article includes a title, a description, and a topic label. Some examples contain mixed signals, such as business-related terms together with country names or conflict-related words. These cases allow us to test whether a model is learning meaningful topic evidence or simply reacting to obvious class-indicative words.

2. Objective

The objective of this project is to train and compare several text classification models on AG News and then analyze the information each model uses to make predictions. We aim to go beyond reporting a single accuracy score. Instead, our goal is to evaluate model performance, interpret model behavior, and test model reliability under modified input conditions.

The project will answer five main research questions. First, how much information is contained in the title compared with the description? Second, how do classical machine learning, neural CNN, and pretrained Transformer models compare on the same task? Third, do the models rely heavily on obvious topic keywords? Fourth, are the models robust to noise such as removed punctuation, typos, source marker removal, or named entity masking? Fifth, are the models confidence scores reliable, especially for ambiguous or incorrectly classified examples?

3. Dataset

We plan to use the AG News topic classification dataset. The classification task is to assign each news article to one of four categories: World, Sports, Business, and Sci/Tech. Each instance contains a title, a description, and a class label. Because the input naturally has two text fields, the dataset is useful for input ablation experiments comparing title only, description only, and title plus description settings.

Before training, we will split the data into training, validation, and test sets. The validation set will be used for model selection and hyperparameter tuning, while the test set will be kept fixed for final evaluation. For TF-IDF models, we will apply standard text preprocessing such as tokenization and feature extraction with unigrams and bigrams. For neural models, we will tokenize text according to the model requirement and limit sequence length consistently across experiments. We will also keep the same data split across all models so that the comparison is fair.

4. Approach

We will compare three models with increasing complexity. The first model is TF-IDF with Logistic Regression. This will serve as a strong classical machine learning baseline and will also provide interpretability because we can inspect highly weighted words and bigrams for each class. The second model is TextCNN, which uses word embeddings and convolution filters to learn local phrase patterns such as stock market, world cup, or software company. This model provides a neural middle ground between bag-of-words methods and Transformers. The third model is DistilBERT, a pretrained Transformer model fine-tuned for AG News classification. DistilBERT is expected to be useful for examples where context matters or where the title and description contain mixed topic signals.

The first experiment will be input ablation. For each model, we will train and evaluate three input formats: title only, description only, and title plus description. This will show whether the title already contains enough information and whether the description improves prediction quality. The second experiment will be model comparison using the same data split and evaluation protocol for all models.

The third part of the approach will analyze what the models rely on. For the Logistic Regression model, we will inspect top weighted TF-IDF features for each class. We will then perform masking experiments on the test set. Random masking will serve as a control condition, while keyword masking will remove class-indicative words selected from training statistics. We will also use attribution-based masking for selected examples by removing words that strongly affect a trained model prediction. These tests will help determine whether the models depend on obvious keywords or use broader context.

We will also conduct robustness analysis by applying controlled input modifications, including source marker removal, punctuation removal, lowercasing, whitespace normalization, typo insertion, and named entity masking. In addition, we will perform a learning curve analysis by training models on different amounts of training data while keeping validation and test sets fixed. Finally, we will analyze model confidence using reliability diagrams, expected calibration error, and qualitative examples of high-confidence wrong predictions.

5. Evaluation

The main evaluation metrics will be accuracy and macro-F1. Accuracy will show overall classification performance, while macro-F1 will treat all four classes equally and reveal whether a model performs poorly on any specific category. We will also report per-class precision, recall, and F1 for World, Sports, Business, and Sci/Tech. Confusion matrices will be used to identify common mistakes, such as Business versus Sci/Tech or Business versus World.

For ablation and robustness experiments, we will compare the performance drop from the original input to the modified input. A large drop after keyword masking would suggest that a model depends strongly on obvious class-indicative words. A smaller drop under typos, source marker removal, or named entity masking would suggest better robustness. For learning curves, we will plot training set size on the x-axis and accuracy or macro-F1 on the y-axis for each model. For confidence analysis, we will use reliability diagrams and Expected Calibration Error to check whether predicted probabilities match actual correctness. We will also manually inspect wrong predictions and categorize failure cases as ambiguous topic, misleading title, keyword trap, entity shortcut, source artifact, label noise, or insufficient context.

The expected outcome of the project is not only a ranking of the models, but a deeper explanation of their behavior. We expect TF-IDF with Logistic Regression to be fast and interpretable, TextCNN to capture useful local phrases, and DistilBERT to perform strongly on contextual and ambiguous cases. The final result will discuss whether the added complexity of neural and Transformer models provides better accuracy, robustness, and reliability compared with the simpler baseline.